Background Shortcut Learning Mitigation for Waterbirds
Week 2 Progress Report

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1 This Week's Work

This week we tried to improve DPE by changing how its classifiers work together. We tested learned weights, a small neural network, and changes to the prototypes. We also built a viewer to compare the mistakes made by DPE, AutoBackSwap, POTER, and ERM on individual images.

2 DPE Experiments

In Week 1, we ran DPE. It first trains a ResNet-50 to extract image features, then keeps that model fixed. On top of it, DPE trains 15 small classifiers, called *heads*. Each head has two *prototypes*: learned reference points representing landbirds and waterbirds. These are feature vectors, not actual bird photos. A head compares an image's features with its prototypes and gives a probability for each class. DPE averages the 15 predictions.

We used the same Waterbirds splits as before: 4,795 training, 1,199 validation, and 5,794 test images. We ran five normal-ERM models (100 training epochs) and two ERM* models (400 epochs with image augmentation). A different random seed gives a separate training run; three of the five normal-ERM runs were reused from earlier experiments. Heads train for 20 epochs, with learning rate 0.001 and batch size 256.

For the learned combinations, we divided validation into 720 images for training heads and 479 for learning how to combine them. We repeated this division three ways, called Splits 0, 1, and 2, retraining the heads and combinations each time. These fits never use test labels. We report *worst-group accuracy* (WGA): the lowest accuracy among the four bird/background groups. Tables average this number across runs; a gain from 89% to 90% is one percentage point (pp). Except for the explicitly labeled paper row in Table 1, all results are from our own runs [2,3,4].

2.1 Combining the 15 Heads

We expected that giving more weight to useful heads might improve DPE. Simple fixed weights did not help: our three-run study gave 89.48% WGA, compared with 89.52% for averaging [1].

We then tried a learned linear combination and a small neural network (MLP, with 15 inputs and 8 hidden units). We also trained the network while randomly hiding 25% of head predictions, keeping at least one per image. This is *prediction dropout*; it encourages the network to work without depending on one head. All heads are available when testing (Fig. 1). Networks train for 400 steps at learning rate 0.01. Table 1 uses weight decay 0.01; 0.001 gave similar findings.

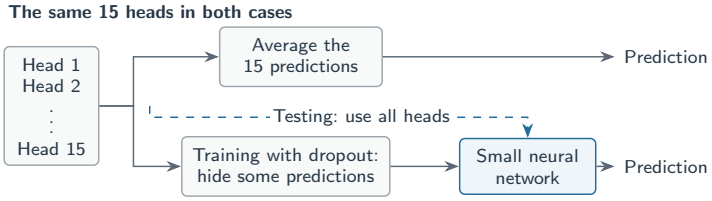


Fig. 1. Two ways to combine the same heads. Ordinary DPE averages their predictions. The learned network can combine them differently. Dropout hides some inputs during training only.

Table 1. Test WGA (%): our results and the original paper, labeled separately. Our means use five normal-ERM runs and two ERM* runs. The variants share our split-validation heads; full-validation reproduction uses all 1,199 validation images.

How predictions are combined	Normal ERM	ERM*
Uniform (our split baseline)	89.23	93.42
Linear combiner (our variant)	89.56	94.00
Neural network (our variant)	89.96	93.76
Dropout network (our variant)	89.85	93.83
DPE (our full-val. reproduction)	89.60	94.08
DPE (original paper [7])	91.00	94.10

In our runs, the network raises normal-ERM WGA from 89.23% to 89.96%. Dropout gives 89.85%, below the plain network. Its gain over averaging also changes from +0.62 to +0.28 to −0.14 pp across the three splits. The result depends on which images train the heads and the network. With only five runs, uncertainty across runs still includes no gain.

On ERM*, both networks improve our split baseline on both runs, but neither beats our 94.08% full-validation reproduction. The paper reports 94.10%; it is a separate reference, not the baseline used to compute our gains. Reserving data for the combiner can offset its benefit.

2.2 Re-pairing and More Prototypes

Table 2. Our prototype-test WGA (%): five normal-ERM runs, two ERM* runs. Ranges cover tested settings; global pooling shows its earlier highest mean. Dashes indicate untested settings, not zero accuracy.

Method	Normal ERM	ERM*
Uniform (our matched split baseline)	89.23	93.42
Re-paired prototypes	89.23	—
Larger prototype subsets (range)	89.23–89.42	93.24–93.42
All 30 prototypes (earlier highest mean)	89.30	—

Main finding: re-pairing changed no predicted labels; larger subsets and global pooling gave only small, inconsistent gains (Table 2) [5,4].

We reused fixed prototypes, testing re-pairings and all 225 pairs at the original scoring scales, plus subsets of 1, 2, 3, 5, or 15 pairs per head (Split 0; Fig. 2).

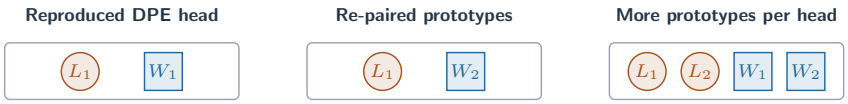


Fig. 2. Re-pairing changes partners; larger subsets add prototypes. Circles/squares represent landbirds/waterbirds; subscripts identify the source head.

2.3 Prototype Weights and a Small Correction

We fitted prototype weights and a *linear correction* using head disagreements. The *confidence adjustment* control changes only the average's confidence and threshold. Fits use 400 steps, learning rate 0.01, and penalties discouraging large changes.

Table 3. Our normal-ERM gains over our matched uniform baseline (pp; five runs). Positive is better. Fixed penalties: 0.01 for prototype weights, 0.1 for confidence adjustment and linear correction. Full ERM* results are retained in the experiment record [4].

Change	Split 0	Split 1	Split 2
Prototype weights	+1.32	−0.02	−1.19
Confidence adjustment	+0.53	+0.18	+0.20
Linear correction	+1.11	+0.40	+0.56

Prototype-weight gains reverse across splits. The linear correction beats both controls on all normal-ERM split means, but loses to confidence adjustment on ERM* (Table 3).

3 Prediction Viewer

We built a local browser viewer [6] containing 13 baseline runs from DPE, AutoBackSwap, POTER, and ERM. It shows shared mistakes, disagreements, and cases one method fixes but another misses. We can filter by species or bird/background group, inspect all predictions for an image, and export selected cases.

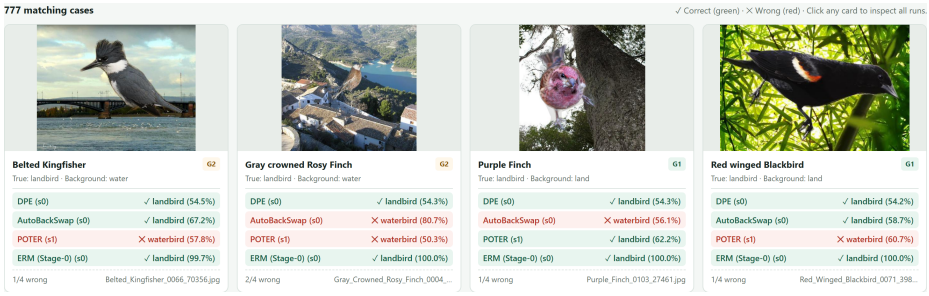


Fig. 3. Screenshot of the disagreement gallery with four baseline methods. Green matches the dataset label; red marks a mistake. These are our earlier baseline reproductions.

In the second card, DPE and ERM are correct while AutoBackSwap and POTER are wrong. The viewer lets us inspect these differences directly.

4 Next Week

- Repeat the linear correction on new runs and splits, comparing averaging and confidence adjustment, including ERM*.
- Use the viewer to inspect errors from re-pairing and larger subsets and changes from new DPE variants. Check why prototype weights depend on the fitting images.
- Follow up re-pairing: test sharper distance-based scores, keeping original pairs as a control, to check whether re-pairing can affect decisions.

These experiments use a test set we have already inspected. Small gains need further checking, and higher WGA alone does not show less reliance on backgrounds.

References

1. Three-run head weighting. Settings and results.
2. Five-run DPE verification. Settings, results and uncertainty.
3. Two-run ERM* combination study. Settings and results.
4. Prototype and correction experiments. Settings and results.
5. Earlier prototype experiments and follow-up. Experiment summary.
6. Waterbirds prediction viewer. Open viewer; usage guide.
7. To et al.: Diverse Prototypical Ensembles Improve Robustness to Subpopulation Shift (2025), Table 2. Original paper.